CAPSTONE • SPRING 2026

Multi-Modal 3D Object Detection

A TransFusion-Based Model for Autonomous Driving Perception

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RESEARCH OVERVIEW

Autonomous driving perception requires accurate 3D object detection using multiple sensor modalities. LiDAR provides precise geometric structure, while cameras contribute rich semantic context. Effectively combining these modalities remains an active research challenge due to differences in representation, alignment, and learning dynamics.

In this work, we implement a TransFusion-based multimodal detection pipeline on the nuScenes dataset and establish a baseline using the mini subset. We then conduct a series of controlled experiments to analyze model behavior across fusion strategies, training configurations, and optimization techniques. The goal is to better understand the practical challenges of multimodal fusion and identify directions for improving robustness and efficiency.

Objective:

To build a working multimodal 3D detection pipeline and systematically evaluate how fusion mechanisms behave under different training and optimization conditions.

SYSTEM ARCHITECTURE

The system is based on the TransFusion framework, which combines LiDAR and camera data using a transformer-based fusion mechanism.

LiDAR data is processed using a voxel-based backbone to extract spatial features, while image data is passed through a convolutional backbone to capture semantic information.

Fusion is performed through cross-attention, where LiDAR-generated object queries selectively attend to relevant image features.

This design allows the model to adaptively integrate information from both modalities instead of relying on fixed fusion rules.

BASELINE PERFORMANCE

- Established a stable baseline for multimodal 3D object detection on the nuScenes mini dataset
- Strong performance on safety-critical classes:
Cars: 37.4% AP
Pedestrians: 43.2% AP
- Moderate performance observed on mid-frequency classes:
Bus, Truck, Barrier → reasonable detection but inconsistent
- Lower performance on less frequent classes:
Bicycle, Motorcycle, Traffic Cone → limited detection capability
- LiDAR-only configuration evaluated for context:
~67% AP for cars and pedestrians (higher due to stronger geometric cues)
- Performance gap attributed to:
Limited sample diversity Underrepresentation of certain object classes Difficulty in learning cross-modal feature alignment
- LiDAR-only models generalize better under these conditions due to direct geometric representation
- Overall, the multimodal baseline remains consistent and interpretable, providing a strong foundation for analyzing fusion behavior across classes

KEY OBSERVATIONS

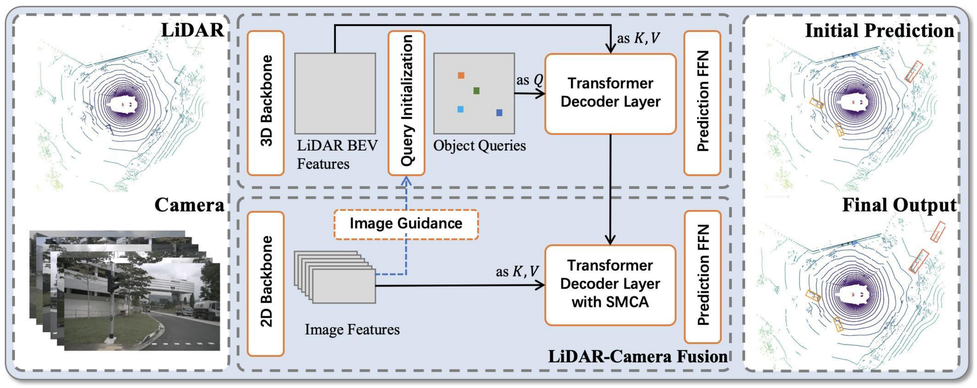
- Performance remains stable across optimization strategies
- Limited gains from loss tuning, distillation, and ensembling
- Strong performance on frequent classes (cars,pedestrians)
- Reduced accuracy on rare object categories
- Fusion behavior depends on feature alignment and representation

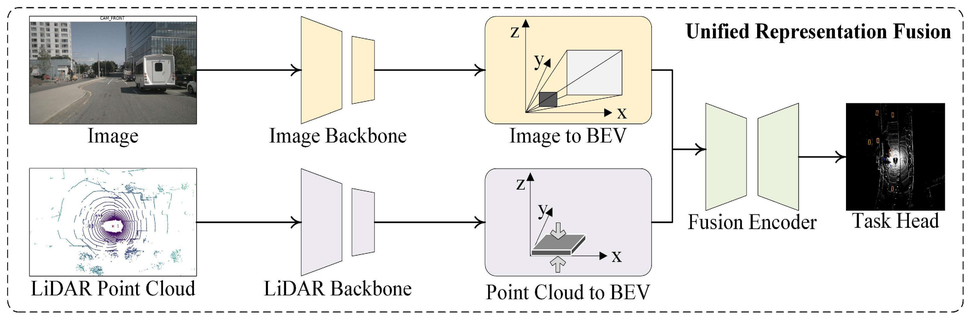
EXPERIMENTS EXPLORED

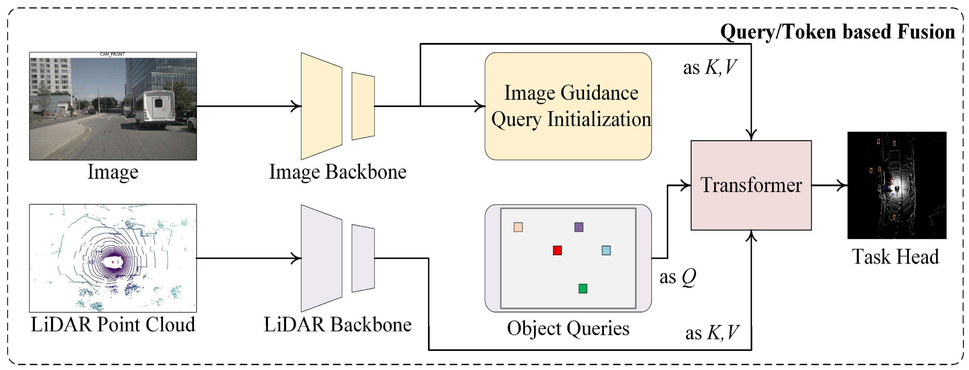
- Conducted a structured study to evaluate multimodal 3D detection across different design choices
- Model Optimization:
Loss rebalancing, knowledge distillation, learning rate tuning
- Data Strategies:
Data augmentation, multi-scale training for improved generalization
- Inference & Efficiency:
FP16 quantization, threshold tuning, model ensembling
- Performance remained stable across all experiments with only marginal variations
- Indicates system behavior is driven by architecture and data representation, rather than isolated optimization techniques
- Highlights the need for deeper improvements in fusion mechanisms and learning dynamics

RESEARCH DIRECTION

- Integrating physics-informed learning to guide multimodal perception
- Incorporating geometric priors
(object size, ground-plane consistency, spatial relationships)
- Exploring spatial reasoning techniques
(BEV representations, geometric feature encoding)
- Improving cross-modal alignment between LiDAR and camera inputs
- Enhancing robustness in challenging scenarios
(sparse data, occlusions, sensor inconsistencies)
- Moving toward geometry-aware, physically grounded perception systems







INSIGHTS

Multimodal detection performance is governed by interactions between data representation, feature alignment, and architectural design. Improvements are not driven by isolated techniques but require integrated changes to fusion and learning mechanisms. These findings highlight the importance of analyzing system behavior holistically rather than focusing on individual components.

CONCLUSION

- Developed a complete TransFusion-based multimodal 3D object detection pipeline
- Established a stable baseline on the nuScenes mini dataset
- Demonstrated strong performance on safety-critical classes (cars and pedestrians), with varying performance across object categories
- Systematic experiments showed limited gains from isolated optimization techniques
- Results highlight the importance of data representation, feature alignment and architectural design in multimodal fusion
- Provides a clear and interpretable foundation for analyzing multimodal detection behavior and guiding future improvements

Future Work

- Conduct full cross-dataset evaluation (e.g., Waymo, KITTI) to test generalization across environments
- Improve performance on rare object classes through targeted data balancing and sampling strategies
- Explore real-time deployment optimizations for reduced latency and improved efficiency
- Perform detailed error analysis to better understand failure cases and guide model improvements

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